Below is a **complete and practical guide** to **Java Command Line Arguments** with examples, and **how to pass them in STS (Spring Tool Suite / Eclipse)**.

✅ 1) What are Java Command Line Arguments?

Command line arguments are values you pass to a Java program when running it from terminal/STS.

They are received in the main() method:

```
public static void main(String[] args) {
}
```

- args is an array of String
- Each value passed becomes one element in args[]

✅ 2) Basic Example

✅ Program: Print arguments

```
public class CmdArgsDemo {
   public static void main(String[] args) {
      System.out.println("Total arguments: " + args.length);

      for (int i = 0; i < args.length; i++) {
         System.out.println("args[" + i + "] = " + args[i]);
      }
   }
}
```

✅ Run from command prompt:

```
java CmdArgsDemo hello world 123
```

✅ Output:

```
Total arguments: 3
args[0] = hello
args[1] = world
args[2] = 123
```

✅ 3) Passing Values with Spaces (Important)

If you want to pass arguments that contain spaces, use quotes.

✅ Example:

```
java CmdArgsDemo "Hello World" "Java Developer"
```

Now:

- args[0] = Hello World
- args[1] = Java Developer

✅ 4) Common Real-World Example: Name, Age, Salary

```
public class EmployeeArgs {
   public static void main(String[] args) {

      String name = args[0];
      int age = Integer.parseInt(args[1]);
      double salary = Double.parseDouble(args[2]);

      System.out.println("Name   : " + name);
      System.out.println("Age    : " + age);
      System.out.println("Salary : " + salary);
   }
}
```

✅ Run:

```
java EmployeeArgs Anurag 23 50000
```

✅ Output:

```
Name   : Anurag
Age    : 23
Salary : 50000.0
```

✅ 5) Best Practice: Validate Arguments

To avoid ArrayIndexOutOfBoundsException, always validate:

```
public class SafeArgs {
   public static void main(String[] args) {
```

```
        if (args.length < 2) {
            System.out.println("Usage: java SafeArgs <username> <password>");
            return;
        }

        String username = args[0];
        String password = args[1];

        System.out.println("Username: " + username);
        System.out.println("Password: " + password);
    }
}
```

✅ 6) Real CLI style: Using Flags (like --name --age)

✅ Example:

```
public class FlagArgs {
    public static void main(String[] args) {

        String name = null;
        int age = 0;

        for (int i = 0; i < args.length; i++) {
            if (args[i].equals("--name")) {
                name = args[i + 1];
            }
            if (args[i].equals("--age")) {
                age = Integer.parseInt(args[i + 1]);
            }
        }

        System.out.println("Name: " + name);
        System.out.println("Age : " + age);
    }
}
```

✅ Run:

```
java FlagArgs --name Anurag --age 23
```

✅ 7) Difference: Java command vs program arguments

When running Java from terminal, there are two types of arguments:

✅ A) JVM Arguments (before class name)

Example:

java -Xmx512m -Denv=prod CmdArgsDemo hello

These include:

- -Xmx512m → memory
- -Denv=prod → system property

✅ B) Program Arguments (after class name)

Example:

java CmdArgsDemo hello

✅ hello is a program argument → goes into args[].

✅ 8) How to Use Command Line Arguments in STS (Spring Tool Suite)

✅ Method 1: Pass Command Line Arguments for a Java Class

Step-by-step:

1. Right click your Java file
2. Select **Run As → Java Application**
3. Then go to:
   Run → Run Configurations...
4. Select your program under **Java Application**
5. Open the **Arguments** tab
6. In **Program arguments**, type:

Example:

Anurag 23 50000

7. Click **Apply**
8. Click **Run**

✅ Output will reflect your passed arguments.

✅ Method 2: Pass Args in Spring Boot Application (STS)

Step-by-step:

1. Right-click your Spring Boot project
2. Run → **Run Configurations**
3. Select:
   Spring Boot App
4. Open **Arguments**
5. Under **Program arguments** add:

Example:

--name=Anurag --age=23

6. Click Run

✅ 9) Example: Read args in Spring Boot

```
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class DemoApplication {

   public static void main(String[] args) {
      System.out.println("Args length = " + args.length);

      for (String a : args) {
         System.out.println(a);
      }

      SpringApplication.run(DemoApplication.class, args);
   }
}
```

Run in STS with:

--server.port=9090 --name=Anurag

✅ 10) STS: JVM Arguments vs Program Arguments (Very Important)

Inside STS Run Configurations:

✅ Program Arguments

Go to:
Arguments tab → Program arguments

Example:

Anurag 23

These go into:

String[] args

✅ JVM Arguments

Go to:
Arguments tab → VM arguments

Example:

-Denv=dev -Xmx512m

These go into:

System.getProperty("env")

✅ 11) Quick Example: Read VM Argument (-D)

Run Config VM Arguments:

-Ddb=mysql

Code:

```
public class VMArgsDemo {
   public static void main(String[] args) {
      String db = System.getProperty("db");
      System.out.println("DB = " + db);
   }
}
```

✅ Output:

DB = mysql

✅ Summary Table

Type	Where to pass in STS	Example	Used In Code
Program arguments	Arguments → Program arguments	Anurag 23	args[]

JVM / VM arguments	Arguments → VM arguments	-Denv=dev	System.getProperty()

If you want, I can also give you a **mini project in Java** (STS ready) that takes:
✅ input file path + output file path as command line arguments and runs like a real tool.